The Causal Model and Notation

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Notation

Consider a typical setting where we have measured some treatment A, some set of pre-treatment variables L, and an outcome Y. We will assume that the data are generated by the following causal model (but alternative definitions can be achieved using other causal models):

$$\begin{aligned} L &= f_L(U_L) \\ A &= f_A(L, U_A) \\ Y &= f_Y(A, L, U_Y) \end{aligned}$$

- The functions f are assumed fixed, but likely unknown.
- $U = (U_L, U_A, U_Y)$ is a vector of exogenous, unmeasured random variables.
- This system is referred to as a Non-parametric Structural Equation Model (NPSEM).
- We will assume that this system of equations exists in nature and is responsible for generating our observed data.

> We will often refer to L as confounders, A as the treatment or exposure, and Y as the outcome.

Take for example the following sample of data:

sex	bmi	age	smoke	trt	event
Male	23.8	56	No	1	1
Male	33.4	80	Former Smoker	0	1
Male	27.5	46	No	0	1
Male	21.9	53	No	1	0
Male	17.2	88	Active Smoker	0	0
Female	28.1	79	Former Smoker	1	0

sex	bmi	age	smoke	trt	event
Male	25.9	50	Active Smoker	1	0
Male	21.4	66	No	0	0
Female	30.9	96	Former Smoker	0	0
Male	24.5	88	Former Smoker	1	1

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Let, L = `sex`, `bmi`, `age`, and `smoke`; A = `trt`; and Y = `event`. When we say that we assume the previous NPSEM, we are positing that:

- the variable `trt` was generated from an unknown function of the random variables `sex`, `bmi`, `age`, and `smoke`, as well as a set of unmeasured random variables not present in our data.
- that the variable `event` was generated from an unknown function of the variables that generated `trt` in addition to `trt` itself.

Note that in some cases, we may know one or more of the functions f. For example, if our data came from a randomized clinical trial for A, then we know the function f_A.

Counterfactuals

Central to how we will define causal effects is the concept of *counterfactual random variables*.

> Counterfactuals are random variables that would have been observed, possibly contrary to fact, in an alternative world.

For example, consider a scenario where we are interested in the value of Y in a hypothetical situation where, instead of the variable A being equal to its observed value, A is set to some other value.

- The value of Y in this hypothetical situation is a counterfactual random variable.
- let $\mathsf{d}(A, L)$ be a function that takes a treatment value A and a covariate profile L and returns a new value of treatment. **We will refer to the function** d **as a *shift* function or a *general hypothetical intervention*.**
- Denote the value of Y in the hypothetical world where treatment is set to the value $\mathsf{d}(A, L)$ as Y^{d}.

Returning to the data example, imagine we are interested in the value of `event` if `trt` was replaced with the output of a function d that always returns 1:

$$
\begin{aligned}
L &= f_L(U_L) \\
A^{\mathsf{d}} &= \mathsf{d}(A, L) = 1 \\
Y^{\mathsf{d}} &= f_Y(1, L, U_Y)
\end{aligned}
$$

Here we introduce some new notation A^{d} to refer to the post-intervention exposure. If we had the ability to collect data from this alternative NPSEM, the data may instead look like this:

sex	bmi	age	smoke	trt	event
Male	32.6	69	No	1	0
Male	30.7	59	Former Smoker	1	0
Female	21.5	61	No	1	0
Male	26.9	49	No	1	0
Male	32	41	No	1	0
Female	30.9	96	Former Smoker	1	0
Female	28.8	81	No	1	0
Female	23.7	55	No	1	0
Male	21	83	Former Smoker	1	0
Male	25.1	62	No	1	0

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Unfortunately, we are never able to collect data from this alternative world. This is called the fundamental problem of causal inference.

The previous NPSEM is the simplest causal model we will assume in this workshop. However, real data is often much more complex and may be characterized by:

- time-varying variables
- loss-to-follow-up

As such, we need to modify and introudce some additional notation:

Symbol	Definition
i	The index (i.e. a row in a dataset) of an observation from a data set with n total units (i.e., the total number of rows)
t	The index of time for a total number of time points τ
L_t	Confounders at time t

Symbol	Definition
A_t	A vector of intervention variables (i..e, treatment or exposure) at time t
Y	An outcome variable observed at the end of the study, that is at time $\tau + 1$. Earlier measures of the outcome can be included in L_t.
C_t	A indicator variable that a unit is *observed* (not censored) at time $t + 1$
$O_1, \ldots, O_n$	A sample of n i.i.d observations with $O = (L_1, A_1, C_1, L_2, A_2, C_2, \ldots, L_\tau, A_\tau, C_\tau, Y)$
$\bar{X}_t = (X_1, \ldots, X_t)$	The history of a variable up until time t
$\underline{X}_t = (X_t, \ldots, X_\tau)$	The future of a variable, including time t
$H_t = (\bar{A}_{t-1}, \bar{L}_t)$	The history of all variables up until just before A_t
ϵ_t	A randomizer
$\mathsf{d}(a_t, h_t, \epsilon_t)$	A function that maps A_t, H_t, and ϵ_t to a new value of treatment A_t^{d}

References

Díaz, Iván, Nicholas Williams, Katherine L Hoffman, and Edward J Schenck. 2023. "Nonparametric Causal Effects Based on Longitudinal Modified Treatment Policies." *Journal of the American Statistical Association* 118 (542): 846–57.